ACI Learning Notes – Part 1 (Core Fabric & Forwarding)

1. ACI Core Components

- **APIC** → Management & Policy
- **SPINE** → COOP Database; → Fabric Transit
- **LEAF** → Endpoint Learning; → Policy Enforcement; → Routing; → VXLAN Encapsulation

Remember:

APIC = Brain
Spine = Directory Service
Leaf = Actual Workhorse

2. Endpoint Learning

Example:

Server
IP : 10.10.10.10
MAC : 0055.1111.1111

Connected to Leaf101
↓
Server appears
↓
Leaf101 learns endpoint
↓
Stored in EPM
↓
Advertised to COOP
↓
Spine updates database

COOP Entry:

10.10.10.10	→ Leaf101
MAC-A	→ Leaf101

3. EPM vs COOP

EPM (On Leaf)

- IP
- MAC
- EPG
- VRF
- Port
- VLAN
- ...

Think: Detailed Endpoint Database

COOP (On Spine)

- IP → Leaf
- MAC → Leaf
- ...

Think: Fabric Directory

Memory Trick:

EPM = HR System | COOP = Company Directory

4. Endpoint Mobility (vMotion)

Before: VM1 10.10.10.10 → Leaf101
vMotion
After: VM1 10.10.10.10 → Leaf102

Changes:

- ✓ IP = Same ✓
- ✓ MAC = Same ✓
- ✓ Leaf Location = Changes

COOP Updates:

OLD:
10.10.10.10 → Leaf101

NEW:
10.10.10.10 → Leaf102

5. Proxy ARP

Question: Who has 10.10.10.20?

Traditional Network:

ARP Flood

ACI:

Leaf already knows
↓
Proxy ARP Reply
↓
Host gets MAC

Important:

Proxy ARP gives
MAC Information
NOT
VTEP Information

6. Host A → Host B Packet Flow

Host A 10.10.10.10 (on Leaf101)
Host B 10.10.10.20 (on Leaf102)

1. ARP Request (Who has 10.10.10.20?)
2. Proxy ARP
3. Host A learns MAC-B
4. Original Packet Sent
5. Leaf101 checks COOP
6. Destination = Leaf102
7. VXLAN Encapsulation
8. Spine → Leaf102 → Host B

7. VXLAN

Host A – Leaf101 → Spine → Leaf102 – Host B

- Host never sees VXLAN.
- Host thinks: MAC-A → MAC-B
- ACI sees: Leaf101 VTEP → Leaf102 VTEP

Memory:

ARP = MAC Discovery
COOP = Endpoint Location
VXLAN = Transport

8. Glean

Occurs when endpoint unknown.

Example: 10.10.10.20 ?

Not in:

- EPM
- COOP

Glean
↓
Endpoint Discovery
↓
Learn Endpoint
or
Drop Packet

Symptoms:

First Ping Fails
Second Ping Works

9. BD vs VRF vs EPG

BD	VRF	EPG
Flood Domain Subnet Gateway	Routing Table	Security Group
Question: Where can I live?	Question: Where can I route?	Question: Who am I?

Contract = Permission
Question: Who can I talk to?

10. Contract Processing

WEB EPG → Traffic → APP EPG

Leaf checks: Contract?

- Yes → Forward
- No → Drop

11. Important Concept

This is usually misunderstood:

ARP Success
↓
DOES NOT mean
↓
Ping Success

You may see:

- ARP ✓
- COOP ✓
- Endpoint ✓
- Ping ✗

Reason: Contract Missing

12. Zoning Rule

GUI: Contract
↓
Leaf Hardware: Zoning Rule

Think:
Contract
↓
Compiled
↓
Zoning Rule
↓
ASIC Enforcement

13. Anycast Gateway

Gateway: 10.10.10.1
Exists on every leaf hosting that BD.

Example: Leaf101 – – – Leaf102 – – – Leaf103
All can answer.

14. L3Out Fundamentals

10.10.10.10 → Traffic → 8.8.8.8

Host ARPs for Default Gateway 10.10.10.1

Host does NOT ARP for 8.8.8.8

Leaf does:

- Route Lookup
- Contract Check
- L3Out
- External Router

15. Golden Rules (Must Memorize)

Rule #1 : Leaf learns endpoints, Spine stores location.
Rule #2 : Proxy ARP = MAC discovery.
Rule #3 : COOP = Endpoint location.
Rule #4 : VXLAN = Transport.
Rule #5 : Routing + Contract = Communication.
Route only ≠ Communication.
Contract only ≠ Communication.
Need BOTH.

Your Current ACI Progress

Fabric Discovery	☑ Good	Endpoint Mobility	☑ Good	L3Out	↻ Learning
EPM/COOP	☑ Good	Contracts	☑ Good	MP-BGP EVPN	✗ Not Started
Proxy ARP	✓ Good	Zoning Rules	☑ Basic	Multi-Pod	✗ Not Started
				Multi-Site	✗ Not Started
				Route Leaking	↻ Learning

Next Focus

- L3Out Packet Flow
- External EPGs
- MP-BGP EVPN Routes